

DRCW-ASEG: One-versus-One distance-based relative competence weighting with adaptive synthetic example generation for multi-class imbalanced datasets

Zhong-Liang Zhang^{a,b,c}, Xing-Gang Luo^{a,b,*}, Sergio González^c, Salvador García^c, Francisco Herrera^{c,d}

^a School of Management, Hangzhou Dianzi University, Hangzhou 310018, China

^b School of Information Science and Engineering, Northeastern University, Shenyang 110819, China

^c Department of Computer Science and Artificial Intelligence, University of Granada, Granada 18071, Spain

^d Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah, Saudi Arabia

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ABSTRACT

Multi-class imbalance learning problems suffering from the different distribution of classes occur in many real-world applications. One-versus-One (OVO) decomposition strategy is a common and useful technique used to address multi-class classification problems, which consists in dividing the original multi-class problem into all binary class sub-problems. The effort to reduce the effect of non-competent classifiers has proven to be a useful way of improving the performance in the OVO scheme. However, these approaches might not be effective for imbalance scenarios, since they are based on standard biased learning procedures. On this account, we propose a novel approach named Distance-based Relative Competence Weighting with Adaptive Synthetic Example Generation (DRCW-ASEG), which properly addresses the synergy between imbalance learning and dynamic classifier weighting in OVO scheme. This new proposed algorithm aims to dynamically produce synthetic examples of minority classes in the stage of dynamic weighting process. We develop a thorough experimental study in order to verify the benefits of the proposed algorithm considering different base binary classifiers.

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1. Introduction

The class imbalance learning problem, also known as imbalanced dataset, refers to the classification task, where one or more classes are under represented in the dataset [1,2]. Learning from datasets with this under-representation results in classifiers biased towards classes with more representatives and loss of prediction of the minority classes [3].

The class imbalance problem has drawn the attention of practitioners over the years, due to its relevance in many real-world classification tasks, such as human activity recognition [4], medical diagnosis [5], data stream analysis [6], and facial expression recognition [7]. In these problems, the goal or the objective class tends to be an under-represented class. However, this attention has been focused mostly on binary class imbalance problems.

Multi-class imbalance problems are considerably more difficult to address than two-class scenarios, since the decision boundary involves distinguishing among more classes. And the vast production for binary imbalance problems unfortunately cannot be directly applied to multi-class imbalance problems [8].

However, multi-class classification problems can be addressed by the usage of binarization decomposition strategies, which aims to divide the complex problem into several simpler two-class problems [9]. Several alternatives following this divide-and-conquer strategy can be found in the specialized literature [10]. Among them, One-versus-One (OVO) [11–13] and One-versus-All (OVA) [14] are the most popular techniques. In this study, we only focus on dealing with multi-class imbalanced datasets using the OVO strategy, as it has proven to perform better than the OVA scheme in this scenario [8,9].

OVO decomposition strategy constructs a system with multiple binary classifiers to discriminate between each pair of classes. For a given test pattern, all the binary classifiers will be triggered. Although, as they are only trained with two-class instances, some of them do not have sufficient knowledge to predict correctly.

* Corresponding author at: School of Management, Hangzhou Dianzi University, 310018 Hangzhou, China.

E-mail addresses: zzl19860210@126.com, zllzhang@hdu.edu.cn (Z.-L. Zhang), xgluo@mail.neu.edu.cn (X.-G. Luo), sergiogvz@decsai.ugr.es (S. González), salvagf@decsai.ugr.es (S. García), herrera@decsai.ugr.es (F. Herrera).

This problem is called as the “non-competent classifiers problem” [15–17].

Recently, some techniques have been proposed to manage this problem for the standard multi-class classification task. Among them, Distance-based Relative Competence Weighting for OVO strategy (DRCW-OVO) [17] has better and more robust performance. In DRCW-OVO, the confidence degrees of the classifiers are weighted depending on their competence associated with the average distances between the classes and the query instance. These average distances are computed with the nearest neighbors of the unlabeled example for each class. The classes closer to the instance to be classified get higher weights in the aggregation method.

However, it may be useless to directly employ DRCW-OVO to handle the class imbalance problems. In the multi-class imbalanced dataset, the minority classes might be much sparser than those of the majority classes within the neighborhood of a given instance to be classified. Therefore, the average distances of the former would be larger, deviating the decision from the minority classes. In this sense, the essential of the drawback for DRCW-OVO to deal with the non-competent classifier problem is the skewed distribution of the local region surrounding the tested example. Therefore, it seems to be reasonable for us to develop an approach to interpolate synthetic examples with the aim of balancing the local region when managing the non-competent classifiers existing in the OVO system for handling multi-class imbalanced datasets.

The goal of this paper is to introduce a new decomposition framework named DRCW-ASEG that is applicable to multi-class imbalanced datasets and empowers OVO decomposition with the DRCW strategy. At the same time, DRCW-ASEG deals with multi-class imbalance problems and with the “non-competent classifiers problem”. This novel approach alleviates the bias towards the majority classes of the post-processing DRCW method, due to the use of the Adaptive Synthetic Example Generation procedure (ASEG) in the stage of dynamic weighting process. We should emphasize that whereas the approaches of adaptively generating minority examples according to their distributions, like ADASYN [18], have been already considered for imbalanced datasets, they have not been considered to manage the non-competent classifiers in OVO scheme. Notice that in this paper, the synthetic examples are generated to balance the local region, and they are not used to establish binary classifiers.

Concretely in DRCW-ASEG, the original multi-class imbalance problem is divided into several two-class sub-problems according to the OVO strategy. And, these are addressed by the binary class imbalanced methods. The non-competent classifier problem is addressed with DRCW but applied to a balanced local region of the sample needed to be classified generated by ASEG. This balanced region is obtained by analyzing the neighborhood of the query example. Then, new synthetic examples are locally generated by the interpolation of those minority classes in the targeted neighborhood.

In order to show the validity of our method, we carry out a thorough experimental study on twenty multi-class imbalanced datasets selected from the KEEL dataset repository [19,20]. Regarding the imbalance learning method in our scenario, six well-known class imbalance learning methods are selected to handle the two-class imbalanced datasets derived from the OVO decomposition, including three basic resampling techniques (Random Under Sampling (RUS), Random Over Sampling (ROS) and SMOTE [21,22]) and three ensemble learning methods (SMOTEBagging [23], RUSBagging [24] and SMOTEBoost [25]).

Macro Average Arithmetic (MAvA) metric [26] is employed as the performance measure and the proper statistical tests suggested in [27] are used to study the significance of the results.

The main contributions of this paper with respect to previous works can be summarized as follows:

- In this paper, we focus on dealing with considerably complicated multi-class imbalance learning tasks using the OVO decomposition technique. We provide the detailed analysis showing that the classic approaches for the management of non-competent classifiers are inefficient in the scenario of multi-class imbalanced datasets.
- In order to alleviate the negative effect of non-competent classifiers using OVO method to handle multi-class imbalanced problems, we develop a new method named DRCW-ASEG which generates synthetic examples by dynamically learning the local information from the data distribution.
- We carry out extensive experiments on real-world datasets to evaluate the performance of the proposed methods. We also study the internal mechanism of the proposed method by analyzing the classification performance on per class.

The rest of this paper is organized as follows. The previous research related to this study, namely the imbalanced classification problem, solutions for multi-class imbalanced datasets, the OVO decomposition scheme and the non-competent classifier problem are introduced in Section 2. Then, in Section 3 we present our methodology for integrating an adaptive synthetic example generation into the OVO scheme, the DRCW-ASEG method. In Section 4, the experimental framework is given, including the datasets, base learner, binary imbalance learning algorithms and measure metric. And then, the complete empirical study and its analysis is carried out in Section 5. Finally, the conclusions and discussions are summarized in Section 6.

2. Background

In this section, we present two scenarios for class imbalanced datasets according to the number of classes. Section 2.1 is devoted to presenting the traditional paradigm of the imbalanced classification problem as a binary scenario. Then, in Section 2.2, we introduce the multi-class imbalanced datasets, its difficulties and possible solutions. Next, we describe OVO decomposition strategy for multi-class classification problems in Section 2.3. Finally, the non-competent classifier problem in OVO scheme is discussed in Section 2.4.

2.1. The imbalanced classification problem

The classification task on a dataset is considered a class imbalance problem, when several classes (minority classes) are underrepresented with respect to the other classes (majority classes), i.e., the number of minority class examples is much less than the number of representatives of the majority classes.

Traditionally practitioners have addressed the problem as a binary scenario dilemma, where the relationship between classes and the goal is well-defined: to reduce the bias towards the majority class and meanwhile, balance the performance on both classes. With this objective, a multitude of approaches have been designed according to four different lines of research: data level methods, algorithmic approaches, cost-sensitive learning solutions, and ensemble learning approaches.

Data level solutions address the origin of the class imbalance problem, i.e. the skewed distribution in the dataset. This is done by sampling the data space to re-balance and reduce the impact of class imbalance [2]. One of the advantages of such a solution is its independence from the classifier used. The two basic ideas of resampling approaches are Random Under/Over-sampling, that randomly remove or duplicate examples from the majority/minority class. Dozens of sophisticated resampling techniques have been proposed over the years. Among them, the Synthetic Minority Oversampling Technique (SMOTE) [21] proposed by Chawla

et al. is the most famous over-sampling technique. SMOTE produces synthetic minority examples by interpolation of the k nearest neighbors of the minority example under consideration. A post-processing SMOTE method using Iterative-Partitioning Filter (IPF) was proposed to alleviate the negative effect from the presence of noisy and borderline examples [28].

Algorithmic adaptation solutions try to adopt appropriate decision threshold to reinforce the learning towards the minority class instances, which are defined as internal approaches in some papers [29–31], since the effect depends on the problems and the classification algorithm [32]. More recent study on this task proposes to use an extension of the nearest neighbor density estimation with a class-wise weighting to deal with the class imbalance problem [33]. Additionally, the influence of global and local paradigms in the weights training phase of Radial Basis Function Networks (RBFNs) for imbalanced datasets is studied in [34].

Cost-sensitive learning considers higher costs for misclassifying the minority class with respect to the majority class, that is, misclassification of minority class is much more expensive. The learning process turns to minimize the cost errors instead of maximization of accuracy rate [35,36].

Finally, ensemble based approaches can be considered as a fundamental area of research. These methods combine the ensemble learning with one of the three previously mentioned strategies in order to balance the performance of both classes and at the same time introduce diversity into the pool of base classifiers [37,38].

2.2. Multi-class imbalanced datasets

In multi-class imbalance classification problems, the canonical solutions may not be effective or achieve a lower performance, since it is easy to loose performance on one class while trying to gain it on others [8]. Relationships among classes are not as clear as in binary scenarios and it is harder to discriminate the minority classes from noise instances, as they are ignored by the classifier. Therefore, it is necessary to develop feasible methodologies to handle these problems. There are some works devoted to handle this task that could be roughly categorized into two groups: straightforward solutions and binarization techniques.

Straightforward solutions have been developed as modifications of the existing algorithm to strengthen their ability to learn from multi-class imbalanced datasets. Among them, AdaBoost.NC [39] with random oversampling is a representative method which combines the negative correlation learning and the boosting procedure to penalize classification errors and encourage the diversity of the ensemble. Static-SMOTE and dynamic-SMOTE [40] are over-sampling techniques used to deal with multi-class imbalance problems, by applying the resampling procedure to each class. A hybrid system is developed by the usage of feature selection based on Binary Particle Swarm Optimization (BPSO) algorithm, combining AdaBoost and k -NN, in which a normalized AUCarea is used to evaluate the fitness of the individual [41]. Additionally, an adapted ensemble learning system attempting to select the best base classifier, ensemble rule, ensemble framework and feature selection method [42] is developed according to the characteristic of the imbalanced data, i.e. imbalance ratio, dimension, and the number of classes. The Mahalanobis Distance-based Over-sampling (MDO) technique is developed in [43], which generates new synthetic instances that have equal Mahalanobis distance from the considered class mean with the candidate samples, while in [44] MDO is embedded into the boosting procedure. In [45], four different types of examples in multi-class datasets are detected: safe, borderline, rare and outliers, whose combinations guide the introduction of synthetic examples. With respect to the cost-sensitive learning, a cost-sensitive boosting algorithm derived from AdaC2 is developed to improve the classification performance for imbalanced

datasets involving multiple classes [46]. In addition, over-sampling, under-sampling and threshold-moving cost-sensitive neural networks are developed in [47] by balancing the cost of misclassification. More recently, a new method named Boosting Adapted for Cost-matrix (BAdaCost) [48] is presented via introducing multi-class cost-sensitive margin and a procedure to estimate the cost matrix from the 0/1-loss confusion matrix.

Binartization techniques, which aim to divide the more complicated original problem into several easier-to-solve binary class sub-problems using the decomposition strategy, have been already proven as effective strategies to deal with multi-class imbalanced problems. A novel decision tree for multi-class imbalance problems is built by using a decomposition scheme and Hellinger distance as the splitting criterion [49]. In [8], the combination of binarization techniques with both preprocessing and cost-sensitive learning is empirically analyzed with respect to several ad hoc approaches. Besides, a method named Near-Bayesian Support Vector Machines (NBSVMs) [50] is proposed to address multi-class imbalanced datasets by the usage of combination of pairwise modified SVMs with boundary shift as well as asymmetric regularization costs using a binarization decomposition scheme. Recently, in [51] multi-class imbalanced data is handled by using ensemble learning techniques for each sub-problem derived from the OVO decomposition strategy.

The most popular binarization techniques are the OVO [11,12] and OVA [14] schemes. The former constructs a system with multiple binary classifiers to discriminate between each pair of classes, while the latter adopts binary classifiers to distinguish each single class from all other classes. It might be inappropriate to employ OVA in problems with skewed distribution, since OVA could introduce an artificial class imbalance [52]. In addition, recent studies [9] clearly show the performance superiority of OVO over OVA methods. Therefore, following this line of research, in this study we are focused on an OVO decomposition scheme to handle multi-class imbalance problems.

2.3. One-versus-One decomposition scheme

The OVO decomposition strategy divides an m -class problem into $\binom{m}{2} = m(m-1)/2$ binary class sub-problems, which are managed by independent classifiers. Each binary model is constructed with a subset of the original training dataset, which only contains the instances of the two corresponding classes. In the prediction phase, a test pattern is classified according to the outputs of each binary classifier, organized as the following score-matrix R :

$$R = \begin{bmatrix} - & r_{12} & \cdots & r_{1m} \\ r_{21} & - & \cdots & r_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & \cdots & \cdots & - \end{bmatrix} \quad (1)$$

where $r_{ij} \in [0, 1]$ is the confidence of the binary classifier distinguishing class i from j , whereas, the confidence in favor of class j is computed as $r_{ji} = 1 - r_{ij}$ (if it is not provided by the classifier). Once the score-matrix is produced, any aggregation methods [9] can be employed to infer the final class. The Weighted Voting (WV) strategy has proven to be one of the best aggregation methods due to its robustness and effectiveness [9,15]. In WV, the final output class is computed over the score-matrix R as shown in the following expression:

$$\text{class} = \arg \max_{i=1, \dots, m} \sum_{1 \leq j \neq i \leq m} r_{ij} \quad (2)$$

2.4. The non-competent classifier problem in the OVO scheme

As we have already mentioned above, there is a well-known drawback, named “the non-competent classifier problem” in the OVO scheme, which must be considered to solve a multi-class problem [15,17]. The classifiers of the OVO scheme are not competent enough to classify all of the classes in the problem, because they are learned only with examples of two classes. However, all the binary classifiers will be fired for a given test pattern, since the competence cannot be known a priori [17], which can lead to incorrect decisions.

Recently, two well-known strategies named DCS-OVO [16] and DRCW-OVO [17] have been proposed which aim to reduce the negative effect of non-competent classifiers. Both of them are based on dynamic post-processing methodology, aiming at modifying the score matrix in the prediction phase. The DRCW-OVO framework has proven to perform better than previous approaches [17]. In DRCW-OVO, the average distances d_i of the nearest neighbors of the query example are computed for each class i during the prediction stage. The relative competence of each binary w_{ij} is estimated as:

$$w_{ij} = \frac{d_j^2}{d_i^2 + d_j^2} \quad (3)$$

Each value r_{ij} of the score-matrix is weighted by the corresponding competence value w_{ij} . Finally, the output class is decided on by the new score-matrix by using the WV method.

These post-processing methodologies are designed within the scope of standard multi-class classification. Therefore, they do not consider the possibility of being biased towards the majority classes in imbalanced scenarios. New methodologies must be designed to take this into account.

3. Distance-based Relative Competence Weighting with Adaptive Synthetic Example Generation (DRCW-ASEG)

In this section, we aim to introduce the framework of DRCW-ASEG for multi-class imbalanced datasets. Firstly, Section 3.1 shows the preliminaries that justify our method. Then, Section 3.2 describes the DRCW-ASEG procedure and explains how the synthetic example generation is performed. Finally, Section 3.3 presents the distance measure metric used in the proposed method.

3.1. Preliminaries for DRCW-ASEG

DRCW-OVO has proven to be an effective methodology for multi-class classification problems. However, it might not be feasible to apply DRCW-OVO to multi-class imbalanced datasets.

After obtaining the outputs of the binary classifiers, DRCW-OVO calculates the average distance between the test pattern and k nearest neighbors of each class. These averages are used to compute the weights to manage the competence of the classifiers. In imbalanced scenarios, the instances of the minority classes might be more sparse. Then, the average distances between the neighbors of these classes and the query instance would be larger than those ones of the majority classes. Therefore, the competence weights would be biased towards the majority classes.

This issue is highlighted in Fig. 1a. Stars, circles and triangles represent three different classes and the square is the query example which needs to be predicted. In this case, $T_1, T_2, T_3, C_1, C_2, C_3$ and S_1, S_2, S_3 are the nearest neighbors ($k=3$) of the test pattern for classes Triangle, Circle and Star, respectively.

As clearly shown in Fig. 1a, the average distance between the test pattern and Star class is shorter than the others, mainly due to the skewed distributions of the classes Circle and Triangle, which

are the minority classes of the problem. This forces the final decision to be biased towards the Star class, that is, the majority class of the problem.

DRCW-ASEG targets this flaw in the following way. Instead of searching for the k nearest neighbors of each class in the original dataset, DRCW-ASEG interpolates synthetic examples for each minority class to force the weighting procedure to remove the bias. We illustrate this idea in Fig. 1b. The hollow examples represent the synthetic examples generated for the Circle and Triangle class, respectively. Therefore, the average distance to each class is closer than before. Using such an operation, the weighting procedure in DRCW-OVO is much fairer for all classes.

3.2. DRCW-ASEG learning algorithm

In this section, we will briefly describe the procedure of synthetic examples generation for DRCW-ASEG. Motivated by the SMOTE [21] and ADASYN [18] algorithms, DRCW-ASEG overcomes the negative influence of skewed class distribution by adaptively interpolating synthetic examples. However, unlike SMOTE and ADASYN, which generate synthetic examples to balance the skewed distribution so that the learning algorithm can work on a balanced dataset, DRCW-ASEG interpolates synthetic examples by adaptively learning local information from the data distribution with the aim of managing the non-competent classifiers in the scenario of multi-class imbalanced datasets. The objective of DRCW-ASEG is to deal with multi-class imbalanced datasets while reducing the different biases derived from the skewed distributions. In order to achieve this goal, there are two aspects to be considered in relation to the impact of these biases.

Firstly, the learning of the multi-class imbalanced dataset must be solved. The multi-class problem is divided into several easier-to-solve two-class sub-problems with the OVO scheme. These sub-problems are addressed by imbalance learning techniques dedicated to binary datasets, which eliminate the main bias produced in the training phase.

Secondly, the non-competent classifier problem of the OVO strategy has to be addressed during the prediction phase using the aggregation method. As previously explained, the identification of the most competent classifiers could also be biased towards those classifiers related to the majority classes. In contrast to the issue first mentioned, this bias must be solved during the prediction phase. In DRCW-ASEG, an adaptive procedure is employed to generate synthetic examples in order to re-balance the local region surrounding the unknown instance. Then, the DRCW procedure can manage the non-competent classifiers in OVO scheme.

The DRCW-ASEG learning algorithm is presented in Algorithm 1. The two mentioned mechanisms which facilitate learning in multi-class imbalanced datasets are explained in detail in this pseudo-code description.

The first mechanism includes Steps 4–9, where the original multi-class imbalanced dataset D_{tr} is decomposed into a series of datasets D_{ij} containing a pair of classes (ω_i, ω_j). Then, the pre-defined binary imbalance learning method H is applied to these two-class imbalanced datasets D_{ij} . After the decomposition process, there could be some binary classifiers responsible for distinguishing each pairwise classes. The final result of this first stage is a pool of classifiers that form the OVO ensemble Θ .

The second mechanism consists of Steps 11–30, where the predictions of the ensemble Θ are aggregated and the competence of the classifiers are managed without any bias coming from the imbalanced class distributions. To achieve this, DRCW-ASEG proposes balancing the local region of the test pattern by using the procedure of adaptive synthetic example generation, which will be explained in Function ASEG (Algorithm 2). Then, the k_1 nearest neighbors for each class ω_i can be selected in the new locally

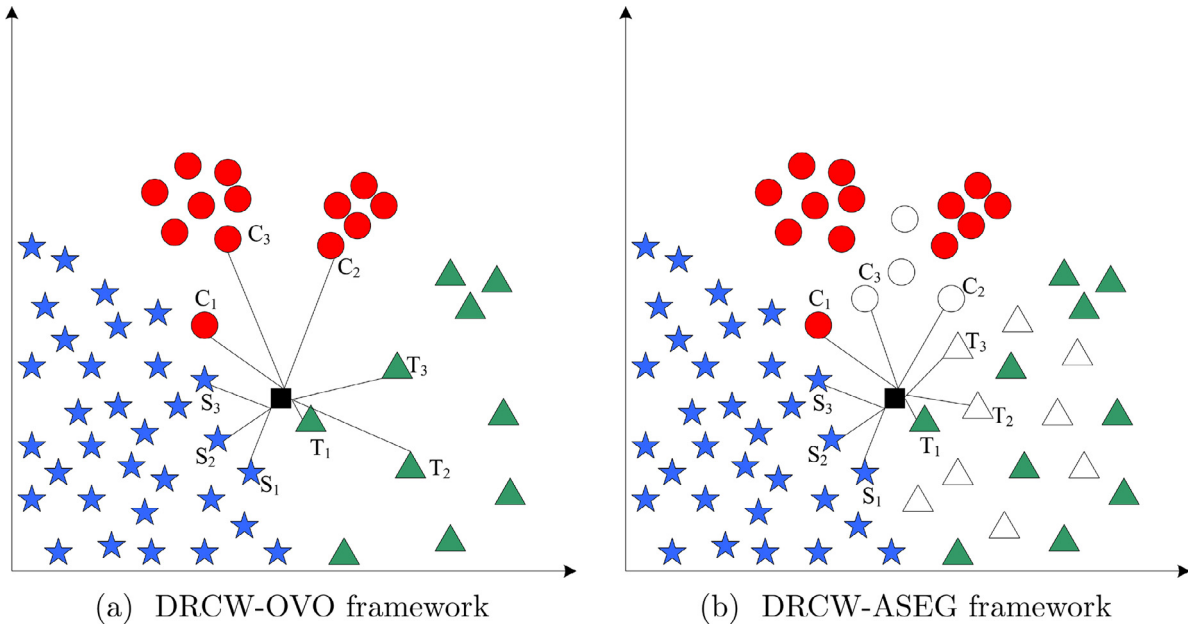


Fig. 1. Potential dilemma when applying DRCW-OVO to imbalanced datasets and DRCW-ASEG solution.

balanced datasets D_{new} . This way, the absence of all bias during the decision of the most competent classifiers is ensured. For each class ω_i , the average of the distances d_i between the query example x_t and its neighbors Ψ^i is calculated. The weights of the classifier competences are computed following the DRCW-OVO approach (Line 27). Finally, these weights are applied to the OVO score-matrix and the final decision is obtained by the class with the maximum weighted vote.

Hereafter, we will describe the procedure of synthetic examples generation for DRCW-ASEG. Motivated by the SMOTE [21] algorithms, DRCW-ASEG overcomes the negative influence of skewed class distribution by interpolating synthetic examples. However, unlike SMOTE, which generates synthetic examples to balance the skewed distribution prior the training phase; DRCW-ASEG interpolates synthetic examples in the local region of the example needed to be predicted by learning from its neighborhood. This operation procedure is done during the prediction of the query instance.

This idea of procedure for interpolating synthetic examples is illustrated in Fig. 2. The highlighted areas surrounded by circles represent the k nearest neighbors search area for each corresponding example. The square represents the test pattern, which is surrounded by $k_2 = 9$ (3 times the number of the classes) nearest neighbors: $S_1, S_2, S_3, S_4, S_5, C_1, C_2, T_1$ and T_2 . It is clear that the circles are the majority class in the local region and the others are the minority classes. The number of each class within the 9-nearest neighbors is $\{5, 2, 2\}$. Then, the ASEG procedure finds the k_3 nearest neighbors ($k_3 = 3$ in this case) of each minority example to decide the number of synthetic examples to be generated for each one. The ASEG procedure is only executed when there is a significantly high Imbalance Ratio (IR) between the largest and the smallest classes, as indicated in Line 16. The degree of significance is set by the parameter $\theta_{threshold}$. In this case, there are 1, 2 and 3 synthetic examples which need to be generated for minority examples T_2, T_1 and C_1 , respectively. It is worth noting that no synthetic examples are generated for minority example C_2 , since there are no circle examples in C_2 nearest neighborhood and it can be considered an isolated case or even noise.

The detailed procedure of ASEG can be found in Algorithm 2. According to the description, the key components of DRCW-ASEG are Steps 6 and 13 in Algorithm 2. The number of the synthetic

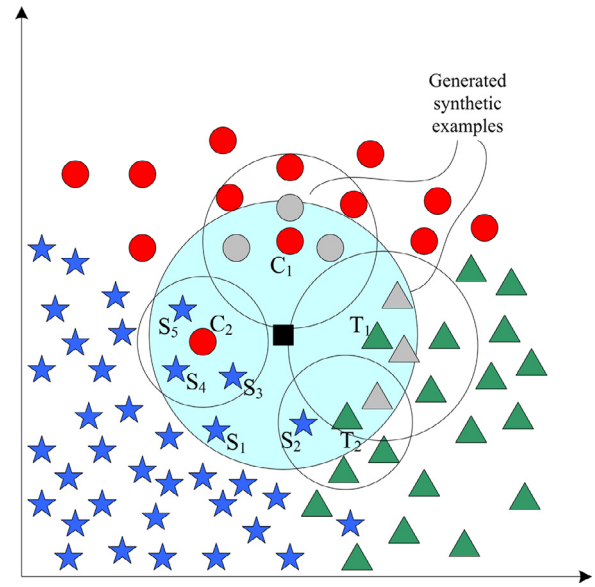


Fig. 2. Synthetic example generation based on DRCW-ASEG.

examples needed to be generated for each minority class in the neighborhood is shown in Step 6. The parameter α defines the re-balance degree desired, where 1 is equally balanced to the majority class. Meanwhile, Step 13 shows the number of the synthetic examples generated for each minority example. This number is proportional to the amount of instances in its classes found in its neighborhood r_i . Once the number of synthetic examples has been selected for each neighbor, the interpolation is done in the same way as SMOTE.

3.3. Distance measure metric

DRCW-ASEG considers the Heterogeneous Value Difference Metric (HVDm) [53] to calculate each distance between the two instances, since this approach allows us to simultaneously handle numerical and nominal attributes unlike the Euclidean distance.

Algorithm 1 Distance-based Relative Competence Weighting with Adaptive Synthetic Example Generation (DRCW-ASEG).

```

1: function DRCW-ASEG (Training dataset  $D_{tr}$ , Testing dataset  $D_{ts}$ ,
   Predefined imbalance learning method for pairwise dataset  $H$ ,
   Number of nearest neighbors used to define the local region of
   the query example  $k_1$ , Number of nearest neighbors in adjusting
   the sampling probability of the minority examples  $k_2$ , Num-
   ber of nearest neighbors used in the dynamic weighting proce-
   dure  $k_3$ , The scaling coefficient  $\alpha$  and the threshold  $\theta_{threshold}$ )
2: Output: the predicted class of each example in  $D_{te}$ 
3: Training phase
4: Classifier ensemble  $\Theta \leftarrow \emptyset$ 
5: for each class pair  $(\omega_i, \omega_j)$  do
6:    $D_{ij} \leftarrow$  a subset of  $D_{tr}$  whose labels are  $\omega_i$  or  $\omega_j$ 
7:   Train a classifier  $C_{ij}$  with  $D_{ij}$  and the learning method  $H$ 
8:    $\Theta \leftarrow \Theta \cup C_{ij}$ 
9: end for
10: Testing phase
11: for each example  $x_t$  in  $D_{te}$  do
12:    $D_{new} \leftarrow D_{tr}$ 
13:   Find  $k_1$  nearest neighbors of the example  $x_t$  in  $D_{tr}$  to ob-
   tain  $D_{neighborhood}$ 
14:   Calculate the degree of class imbalance  $IR = N_s/N_l$ 
15:   where  $IR \in [0, 1]$ ,  $N_s$  is the number of smallest class ex-
   amples,  $N_l$  is the number of largest class examples
16:   if  $IR < \theta_{threshold}$  then
17:      $D_{syn} \leftarrow ASEG(D_{tr}, x_t, D_{neighborhood}, k_2, \alpha)$ 
18:   end if
19:    $D_{new} \leftarrow D_{new} \cup D_{syn}$ 
20:   for each class pair  $(\omega_i, \omega_j)$  do
21:      $r_{ij} = \Theta_{ij}^{x_t}$ 
22:   end for
23:   for each class  $\omega_i$  do
24:     Find  $\Psi^i$  as the  $k_3$  nearest neighbors of the example  $x_t$ 
   in  $D_{new}$ 
25:     Calculate the average distance  $d_i$  between  $x_t$  and  $\Psi^i$ 
26:   end for
27:   Calculate the weights  $w_{ij} = \frac{d_j^2}{d_i^2 + d_j^2}$ 
28:   Obtain the new score matrix  $r_{ij}^{new} = r_{ij} w_{ij}$ 
29:   Obtain the final output  $\varphi(x_t)$  according to the selected
   aggregation strategy
30: end for
31: end function

```

This metric is applied in both the distances of the nearest neighbors of the DRCW procedure and in the ASEG method. The distance between two input examples $\mathbf{x}$ and $\mathbf{y}$ is defined in HVDM method as follows:

$$d_{HVDM}(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{a=1}^f d_a^2(x, y)} \quad (4)$$

where f denotes the number of attributes. The function $d_a(x, y)$ is the distance metric between two values x and y for attribute a and is calculated using the following equation:

$$d_a(x, y) = \begin{cases} 1, & \text{if } x \text{ or } y \text{ is unknown; otherwise...} \\ \text{normalized_vdm}_a(x, y), & \text{if } a \text{ is nominal} \\ \text{normalized_diff}_a(x, y), & \text{if } a \text{ is continuous} \end{cases} \quad (5)$$

Hence, one of the two functions is selected to calculate the distance $d_a(x, y)$ between the values x and y of the attribute a depending on the its type (whether it is nominal or numerical). The

Algorithm 2 ASEG: Adaptive Synthetic Examples Generation.

```

1: function ASEG(Training data  $D_{tr}$ , Testing example  $x_t$ , The lo-
   cal region of the query example  $D_{neighborhood}$ , Number of nearest
   neighbors in adjusting the sampling probability of the minority
   examples  $k_2$ , The scaling coefficient  $\alpha$ )
2: Output: synthetic examples  $D_{syn}$ 
3:  $D_{syn} \leftarrow \emptyset$ 
4: for each class  $\omega_l$  in  $D_{neighborhood}$  do
5:   Calculate the number of synthetic examples:
6:    $G = (N_l - N_{\omega_l}) \times \alpha$ 
7:   where  $N_{\omega_l}$  is the number of class  $\omega_l$  examples in
    $D_{neighborhood}$  and  $N_l$  is the number of largest class examples
8:   for each example  $x_i$  with class  $\omega_l$  in  $D_{neighborhood}$  do
9:     Find  $k_2$  nearest neighbors in  $D_{tr}$  to obtain  $D'_{neighborhood}$ 
10:    Calculate the ratio  $r_i = N'_{\omega_l}/k_2$ 
11:    where  $N'_{\omega_l}$  is the number of class  $\omega_l$  examples in
    $D'_{neighborhood}$ 
12:     $\hat{r}_i = r_i / \sum_{i=1}^{N_{min}} r_i$ 
13:     $g_i = \lceil G \times \hat{r}_i \rceil$ 
14:  end for
15:  for  $j = 1$  to  $g_i$  do
16:    Generate synthetic example  $SynExample$  like SMOTE
17:     $D_{syn} \leftarrow D_{syn} \cup SynExample$ 
18:  end for
19: end for
20: end function

```

distance for nominal attributes is defined as follows:

$$\text{normalized_vdm}_a(x, y) = \sqrt{\sum_{i=1}^m \left| \frac{N_{a,x,\omega_i}}{N_{a,x}} - \frac{N_{a,y,\omega_i}}{N_{a,y}} \right|} \quad (6)$$

where m is the number of classes, $N_{a,x}$ is the number of examples in the training dataset with value x for attribute a , and N_{a,x,ω_i} is the number of examples in the training dataset with value x for attribute a which is labeled as class ω_i .

The distance for continuous attributes is defined as following function:

$$\text{normalized_diff}_a(x, y) = \frac{|x - y|}{4 \cdot \sigma_a} \quad (7)$$

where σ_a is the standard deviation of the values of attribute a in the training dataset.

4. Experimental framework

In this section, we present the set-up of the experimental framework used in the following experiments. Twenty datasets from the KEEL repository [19,20] are selected to analyze the algorithms. The properties of the datasets are presented in Table 1. For each dataset, the number of examples (#Ex.), the number of attributes (#Atts.), the number of numerical (#Num) and nominal (#Nom) attributes, the number of classes (#Cl.), the distribution of class (#Dc) and IR are shown. As for the missing value instances in the datasets (Automobile, Cleveland and Dermatology), we removed them before doing the partitions.

The results are obtained by means of three times 5-fold stratified cross-validation (5-SCV). That is, each dataset is split into 5 folds and each fold contains 20% of the instances of the dataset. For each fold, the algorithm is trained with the instances contained in the remaining folds (80% of the instances of the dataset) and then tested by the current fold. The reason why 5-fold SCV is more appropriate than a 10-fold SCV in imbalance problems was explain in [54]. We use the same data partitions as in the KEEL dataset

Table 1
Summary description of the datasets.

id	Dataset	#Ex.	#Atts.	#Num.	#Nom.	#Cl.	#Dc.	IR
Aut	Automobile	159	25	15	10	6	46/29/13/48/20/3	16
Bal	Balance	625	4	4	0	3	288/49/288	5.88
Car	Car	1728	6	0	6	4	384/69/1210/65	18.62
Cle	Cleveland	297	13	5	8	5	160/54/35/35/13	12.31
Con	Contraceptive	1473	9	6	3	3	629/333/511	1.89
Der	Dermatology	358	34	1	33	6	111/60/71/48/48/20	5.55
Eco	Ecoli	336	7	7	0	8	143/77/2/2/35/20/5/52	71.5
Fla	Flare	1066	11	0	11	6	147/211/239/95/43/331	7.7
Gla	Glass	214	9	9	0	6	70/76/17/29/13/9	8.44
Hay	Hayes-roth	160	4	4	0	3	65/64/31	2.1
Led	Led7digit	500	7	0	7	10	45/37/51/57/52/52/47/57/53/49	1.54
Lym	Lymphography	148	18	3	15	4	4/61/81/2	40.5
New	New-thyroid	215	5	5	0	3	150/35/30	5
Pag	Page-blocks	548	10	10	0	5	491/33/3/9/12	163.67
Pen	Penbased	1100	16	16	0	10	115/114/114/106/114/106/105/115/105/106	1.1
Tae	Tae	151	5	4	1	3	49/50/52	1.06
Veh	Vehicle	846	18	18	0	4	218/212/217/199	1.1
Win	Wine	178	13	13	0	3	59/71/48	1.48
Yea	Yeast	1484	8	8	0	10	463/5/35/44/51/163/244/429/20/30	92.6
Zoo	Zoo	101	16	0	16	7	41/20/5/13/4/8/10	10.25

Table 2
Approaches considered to address the binary class dataset derived from the OVO schemes.

Abbreviation	Method	Short description
RUS	Random Under Sampling	Random elimination of majority class examples
ROS	Random Over Sampling	Random replication of minority class examples
SMOTE	Synthetic Minority Oversampling Technique [21]	Interpolation of new minority class examples
SBA	SMOTEBagging [23]	Bagging with over-sampling with SMOTE
RBO	UnderBagging [24]	Bagging with random under-sampling
SBO	SMOTEBoost [25]	AdaBoost.M2 with over-sampling with SMOTE

Table 3
Parameters considered for the compared methods.

Algorithm	Parameters
DRCW-ASEG	$k_1 = 5$, $k_2 = 3$ times the number of classes, $k_3 = 10$ $\theta_{threshold} = 0.75$ $\alpha \in [0.1, 0.3, 0.5, 0.7, 1.0, 3.0, 5.0]$ – best selected for each dataset
DRCW-OVO	$k = 3$ times the number of classes
OVO-Easy	EasyEnsemble [56] is used as the binary learning method in OVO decomposition scheme The number of subsets is 4 The number of iteration in each AdaBoost ensemble is 10
AdaBoost.NC	ROS is used as the imbalance learning method $\lambda = 2$
Static-SMOTE	$p = 100$ generates 100% new synthetic examples of the selected class in each iteration

repository [19], so that the experimental study can be reproduced by any interested readers.

In order to test our approach and compare the different OVO frameworks, we must choose a variety of methods for two-class imbalanced problems. Table 2 summarizes the considered methods. These classifiers are employed to form the OVO ensemble and independently address each two-class sub-problem. With this objective, we have selected several well-known approaches to deal with binary class imbalance learning problems. Focusing on pre-processing, we have considered three simpler sampling approaches and other three more complex ensemble-based methods. We have used Classification and Regression Tree (CART) [55] as the baseline classifier, where pruning is used and impure nodes must have ten or more examples to be split.

Ensemble-based approaches have been executed with 40 iterations. The number of nearest neighbors considered in SMOTE for each approach was set to 5. Table 3 recalls the parameter settings of our proposal DRCW-ASEG and the initial method DRCW-OVO [17], as well as the state-of-the-art approaches, including OVO-Easy [51], AdaBoost.NC [39] and Static-SMOTE [40], which are employed as benchmark methods.

As mentioned, we have considered Macro Average Arithmetic (MAvA) [26] as the performance measure metric. The MAvA gives the same weight to each class, which is defined as the arithmetic average of the partial accuracy of each class independently.

$$MAvA = \frac{1}{m} \sum_{i=1}^m ACC_i \quad (8)$$

where ACC_i stands for the accuracy rate of the class ω_i .

5. Results and analysis

In this section, we aim to verify the usefulness and robustness of DRCW-ASEG for managing the non-competent classifier problem, while dealing with the multi-class imbalanced datasets. In Section 5.1, we carry out a pairwise comparative analysis of our approach versus the DRCW-OVO method with different well-known imbalance learning methods to solve the binary sub-problems derived from the OVO decomposition. Then, we compare our proposal DRCW-ASEG with the state-of-the-art methods in Section 5.2. Finally, we present the performance of our DRCW-ASEG on each class, comparing the benchmark method DRCW-OVO

Table 4

Complete MAVA test results for DRCW-OVO and our proposal DRCW-ASEG with different binary imbalance learning methods. The best results are highlighted in bold-face.

Dataset	RUS		ROS		SMOTE		SBA		RBO		SBO	
	DRCW-OVO	DRCW-ASEG	DRCW-OVO	DRCW-ASEG	DRCW-OVO	DRCW-ASEG	DRCW-OVO	DRCW-ASEG	DRCW-OVO	DRCW-ASEG	DRCW-OVO	DRCW-ASEG
Aut	73.77	73.80	84.84	83.74	83.47	84.03	83.42	84.37	68.42	72.16	78.85	85.17
Bal	63.05	60.07	60.04	59.94	57.55	57.66	57.92	56.52	63.09	62.58	60.57	57.76
Car	96.51	96.51	96.95	96.50	95.92	96.05	96.06	95.89	96.42	96.50	97.86	97.64
Cle	29.45	31.26	28.66	31.11	28.45	31.02	28.72	28.99	34.38	32.84	29.70	30.22
Con	51.11	50.77	49.29	49.51	49.76	49.82	51.14	50.24	49.69	49.02	49.53	48.82
Der	97.13	97.16	97.65	97.98	97.09	97.71	97.09	97.61	96.53	96.85	96.32	96.95
Eco	69.73	71.89	71.16	71.92	71.67	72.11	71.01	71.28	60.26	64.44	76.42	76.72
Fla	62.27	64.45	59.60	62.46	59.59	60.35	57.68	57.64	63.29	63.49	60.30	60.91
Gla	70.57	75.70	74.96	72.34	71.50	73.84	70.89	73.77	74.05	76.65	75.59	78.32
Hay	85.09	88.16	85.09	88.16	85.09	88.68	85.60	87.65	85.30	88.16	83.08	87.65
Led	68.69	66.87	68.82	66.24	69.80	67.29	67.07	63.26	71.91	70.31	70.72	69.63
Lym	74.32	80.04	79.59	81.36	79.84	80.75	81.76	81.94	66.60	65.95	86.00	88.43
New	93.80	96.10	92.93	95.04	94.04	95.84	93.81	94.92	94.14	95.04	92.22	94.89
Pag	71.60	85.70	73.04	75.39	76.28	80.83	76.30	78.97	73.53	84.76	79.76	81.70
Pen	95.99	96.35	95.93	96.22	96.02	96.22	96.14	96.38	96.77	97.84	96.92	97.67
Tae	58.89	60.68	58.82	59.49	59.51	59.54	56.64	57.33	59.09	61.03	59.56	61.05
Veh	71.92	72.94	72.18	72.54	72.31	72.76	72.33	72.51	75.32	76.29	76.61	76.50
Win	92.53	94.93	95.71	95.35	95.33	95.58	96.45	96.45	96.60	96.75	96.36	97.24
Yea	60.13	58.64	57.40	56.95	58.77	59.09	59.33	59.56	53.45	54.52	58.85	58.79
Zoo	80.48	89.33	88.45	92.26	87.98	92.82	87.98	91.31	79.37	83.10	92.50	92.50
Avg.	73.35	75.57	74.56	75.22	74.50	75.60	74.37	74.83	72.91	74.41	75.88	76.93

Table 5

Results of the Wilcoxon signed-rank test for comparison between DRCW-ASEG and DRCW-OVO with the considered binary imbalance learning methods. A '*' near the p -value means that there are statistical differences with $\alpha = 0.05$ (95% confidence) and a '-' with $\alpha = 0.1$ (90% confidence).

DRCW-ASEG vs. DRCW-OVO	R^+	R^-	Hypothesis	p -value
RUS	173.00	37.00	Rejected for DRCW-ASEG at 5 %	0.011129*
ROS	145.00	65.00	Not rejected	0.135357
SMOTE	194.00	16.00	Rejected for DRCW-ASEG at 5 %	0.000892*
SBA	157.50	52.50	Rejected for DRCW-ASEG at 10 %	0.053406-
RBO	167.00	43.00	Rejected for DRCW-ASEG at 5 %	0.020633*
SBO	161.50	48.50	Rejected for DRCW-ASEG at 5 %	0.032938*

in order to obtain the inherent mechanism of the proposed method in Section 5.3.

5.1. Analysis of the effectiveness of DRCW-ASEG

In this section, our goal is to analyze the effectiveness of synthetic example generation for dynamic classifier weighting to eliminate the biases towards majority classes in the final decisions. For this purpose, six imbalance learning methods, i.e. RUS, ROS, SMOTE, SBA, RBO and SBO, are considered to address the imbalanced binary class datasets derived from the OVO decomposition scheme. Then, DRCW-OVO and DRCW-ASEG are used to obtain the final outputs, with the purpose of proving a significant improvement of DRCW-ASEG as compared to the previous one.

Table 4 gathers the MAVA results for the 20 datasets for the two different OVO frameworks in combination with the 6 binary classifiers selected. The best results have been stressed in bold-face. Overall, these results demonstrate the quality of our proposal DRCW-ASEG as opposed to the previous DRCW-OVO.

On average, DRCW-ASEG obtains better results with each of the six different tested methods than DRCW-OVO. Additionally, for each binary classifier tested, DRCW-ASEG has a greater number of datasets with the best results obtained. That is, for each pair of columns in the results table DRCW-ASEG wins in many more datasets.

In this sense, the greater difference in number of wins is found in SMOTE+Cart as a base learner with 19 best results for DRCW-ASEG and only 1 for the previous approach. In contrast, the smallest difference is found in the combination ROS+Cart with 13 best

Table 6

Complete MAVA test results for DRCW-ASEG and state-of-the-art methods. The best results are highlighted in bold-face.

Dataset	OVO-Easy	AdaBoost.NC	Static-SMOTE	DRCW-ASEG
Aut	34.33	76.32	73.21	85.17
Bal	71.35	63.89	58.44	57.76
Car	98.09	96.75	94.23	97.64
Cle	32.03	31.97	30.96	30.22
Con	51.47	51.20	47.07	48.82
Der	94.23	95.96	91.47	96.95
Eco	43.76	65.80	65.92	76.72
Fla	63.87	57.12	62.30	60.91
Gla	69.52	72.95	70.11	78.32
Hay	86.31	86.98	83.87	87.65
Led	70.62	51.07	70.01	69.63
Lym	34.38	82.20	84.03	88.43
New	95.10	93.77	89.92	94.89
Pag	56.56	67.58	79.19	81.70
Pen	95.33	92.32	87.59	97.67
Tae	62.13	48.64	54.92	61.05
Veh	77.22	72.37	70.91	76.50
Win	94.31	95.06	91.05	97.24
Yea	37.86	54.39	50.02	58.79
Zoo	50.00	90.83	90.00	92.50
Avg.	65.92	72.36	72.26	76.93

results for DRCW-ASEG as compared to 7 for DRCW-OVO, the former still performing better on average.

Independent of the base classifiers, DRCW-ASEG performs better than DRCW-OVO in the majority of datasets. For 9 unique datasets, DRCW-ASEG always achieves better performance for each binary classifier. Meanwhile, DRCW-OVO is globally better in only one dataset (Led).

Table 7

Results of the Wilcoxon signed-rank test for the comparison of DRCW-ASEG and state-of-the-art methods. A ‘**’ near the p -value means that there are statistical differences with $\alpha = 0.05$ (95% confidence) and a ‘-’ with $\alpha = 0.1$ (90% confidence).

Comparison	R^+	R^-	Hypothesis	p -value
DRCW-ASEG vs. OVO-Easy	153.00	57.00	Rejected for DRCW-ASEG at 10 %	0.073138-
DRCW-ASEG vs. AdaBoost.NC	182.00	28.00	Rejected for DRCW-ASEG at 5 %	0.004045*
DRCW-ASEG vs. Static-SMOTE	200.0	10.00	Rejected for DRCW-ASEG at 5 %	0.000390*

Table 8

Number of examples of each class in each dataset and classification performance on each class for DRCW-OVO and DRCW-ASEG (DRCW-OVO, DRCW-ASEG) when SBO is used as the binary learner.

Dataset	Class 1	Class 2	Class 3	Class 4	Class 5
Aut	46 (100.00, 100.00)	29 (61.67, 70.00)	13 (87.78, 87.78)	48 (91.85, 92.59)	20 (67.33, 81.78)
Bal	288 (92.49, 88.34)	49 (0.00, 0.00)	288 (89.23, 84.94)	–	–
Car	384 (97.80, 97.99)	69 (97.66, 97.14)	1210 (100.00, 97.95)	65 (95.97, 97.47)	–
Cle	160 (90.25, 83.07)	54 (13.03, 14.42)	35 (27.14, 26.79)	35 (18.10, 19.05)	13 (0.00, 7.78)
Con	629 (62.70, 62.75)	333 (38.15, 38.25)	511 (47.75, 45.47)	–	–
Der	111 (100.00, 100.00)	60 (87.78, 89.44)	71 (98.57, 98.57)	48 (91.56, 93.70)	48 (100.00, 100.00)
Eco	143 (96.26, 95.79)	77 (78.86, 78.89)	2 (60.00, 60.00)	2 (60.00, 60.00)	35 (60.00, 60.95)
Fla	147 (100.00, 100.00)	211 (51.32, 47.96)	239 (70.77, 73.60)	95 (98.64, 98.64)	43 (26.32, 26.67)
Gla	70 (90.48, 88.10)	76 (69.75, 72.36)	17 (26.67, 43.89)	29 (80.00, 80.00)	13 (100.00, 100.00)
Hay	65 (66.15, 85.13)	64 (83.08, 77.82)	31 (100.00, 100.00)	–	–
Led	45 (64.44, 68.89)	37 (74.52, 73.81)	51 (73.09, 66.42)	57 (63.48, 68.48)	52 (85.58, 90.55)
Lym	4 (100.00, 100.00)	61 (87.92, 88.33)	81 (76.07, 85.38)	2 (80.00, 80.00)	–
New	150 (100.00, 99.11)	35 (93.33, 100.00)	30 (83.33, 85.56)	–	–
Pag	491 (99.12, 98.64)	33 (76.35, 84.29)	3 (100.00, 100.00)	9 (90.00, 80.00)	12 (33.33, 45.56)
Pen	115 (97.68, 97.39)	114 (87.17, 92.12)	114 (100.00, 99.42)	106 (99.05, 99.05)	114 (98.84, 98.26)
Tae	49 (58.67, 59.26)	50 (54.67, 57.33)	52 (65.33, 66.55)	–	–
Veh	218 (97.17, 97.50)	212 (57.54, 55.39)	217 (97.86, 97.55)	199 (53.86, 55.58)	–
Win	59 (100.00, 100.00)	71 (89.81, 91.71)	48 (99.26, 100.00)	–	–
Yea	463 (62.83, 63.13)	5 (100.00, 100.00)	35 (57.14, 58.10)	44 (82.69, 85.19)	51 (41.82, 42.42)
Zoo	41 (97.50, 97.50)	20 (100.00, 100.00)	5 (60.00, 60.00)	13 (100.00, 100.00)	4 (100.00, 100.00)
Dataset	Class 6	Class 7	Class 8	Class 9	Class 10
Aut	3 (64.44, 78.89)	–	–	–	–
Bal	–	–	–	–	–
Car	–	–	–	–	–
Cle	–	–	–	–	–
Con	–	–	–	–	–
Der	20 (100.00, 100.00)	–	–	–	–
Eco	20 (80.00, 80.00)	5 (100.00, 100.00)	52 (76.24, 78.12)	–	–
Fla	331 (14.72, 18.61)	–	–	–	–
Gla	9 (86.67, 85.56)	–	–	–	–
Hay	–	–	–	–	–
Led	52 (76.06, 76.73)	47 (74.44, 65.78)	57 (70.45, 69.90)	53 (63.82, 61.94)	49 (61.33, 53.78)
Lym	–	–	–	–	–
New	–	–	–	–	–
Pag	–	–	–	–	–
Pen	106 (98.10, 97.78)	105 (100.00, 100.00)	115 (96.23, 97.68)	105 (97.14, 98.10)	106 (94.96, 96.87)
Tae	–	–	–	–	–
Veh	–	–	–	–	–
Win	–	–	–	–	–
Yea	163 (81.84, 80.81)	244 (55.82, 54.32)	429 (56.34, 53.93)	20 (50.00, 50.00)	30 (0.00, 0.00)
Zoo	8 (100.00, 100.00)	10 (90.00, 90.00)	–	–	–

For both approaches, SMOTEBoosting is the best base learner in terms of performance. For this combination, DRCW-ASEG is also notoriously better.

In order to establish the significance of the quality of DRCW, we have carried out a statistical analysis with non-parametric hypothesis testing [27]. Concretely, Wilcoxon signed-rank test is employed in the pairwise comparisons of the results for each base learner. Table 5 gathers the ranks and p -values of each equivalence hypothesis, where R^+ represents the sum of the ranks for DRCW-ASEG and R^- is the sum of the ranks for DRCW-OVO.

These final results still indicate the quality of DRCW-ASEG. The ranks obtained are, in the majority, great enough to indicate an improvement in DRCW-ASEG. Additionally, this improvement is reaffirmed for 4 of the 6 tested base learners, where the equivalence hypotheses are rejected with very low p -values, with the remaining two considered low values (0.053406 and 0.135357). Finally, this hypothesis is also rejected for the better resulting combination of both the OVO scheme and SMOTEBoosting.

5.2. Comparison of DRCW-ASEG with state-of-the-art methods

There are a number of approaches already proposed in the literature for handling multi-class imbalanced problems. In this section, we carry out a comparison to analyze the performance of our proposed method DRCW-ASEG with respect to the state-of-the-art methods. All this considered, we have selected three popular methods, i.e. OVO-Easy [51], AdaBoost.NC [39] and Static-SMOTE [40], which are considered the most popular when addressing the multi-class imbalance problem.

The complete results for the examined methods are depicted in Table 6, where the best result for each dataset is highlighted in bold-face. Although OVO-Easy returns the worst average result from all of the examined methods, it achieves the best results in nine out of 20 datasets. OVO-Easy performs much worse in several datasets, i.e. Aut, Eco, Lym and Zoo. This can be observed for cases with some underrepresented classes. Moreover, the average results of AdaBoost.NC and Static-SMOTE are close. However,

AdaBoost.NC outperforms Static-SMOTE in most datasets tested. Finally, these three popular methods perform worse than our DRCW-ASEG, including both the average performance measures over all the tested datasets and the number of datasets in which DRCW-ASEG stands out. This allows us to conclude that the proposed DRCW-ASEG could be a highly efficient solution to the issue at hand, being able to deliver generally improved performance when compared to popular solutions from the literature.

To gain an additional insight into the performance of the examined methods, we have carried out a pairwise statistical analysis by using a Wilcoxon signed-rank test, outcomes of which are presented in Table 7.

The results of the statistical tests shown in Table 7 are clear. For all these tests, we conduct the comparison of DRCW-ASEG and the state-of-the-art technique, showing the sum of the ranks for DRCW-ASEG in R^+ and the sum of the ranks for other methods in R^- . Obtained ranks and p -values show that the proposed method DRCW-ASEG is significantly better than the well-known techniques proposed in the other literature for addressing skewed distributions in multiple class, since all the p -values returned by the Wilcoxon test are lower than our α -value (0.1). In conclusion it must be noted that the proposed method DRCW-ASEG can outperform state-of-the-art methods in multi-class imbalanced datasets, thus being a worthwhile choice for such problems.

5.3. Analysis of the classification performance per class of DRCW-ASEG

The effectiveness of the proposed method DRCW-ASEG in addressing multi-class imbalanced datasets has been justified in the previous sections. The aim of this subsection is to present the internal mechanism that increases the classification performance for multi-class imbalanced datasets. On this account, we want to analyze the classification performance of the proposed method DRCW-ASEG on each class so as to demonstrate the suitability of the mechanism of DRCW-ASEG. With this objective, we have computed the classification accuracy per class of the proposed method DRCW-ASEG and the benchmark method DRCW-OVO for each dataset, which is shown in Table 8.

The results obtained from Table 8 can be summarized as follows:

- (1) The individual results for each dataset show that usually DRCW-ASEG achieves the best results for most of the minority classes. It is important to note, for example, the results in datasets such as *New-thyroid*, in which DRCW-ASEG obtains improvements in classifying the minority classes, i.e. class 2 and class 3, respectively, whereas it achieves slightly worse performance on the majority class (class 1).
- (2) There are some datasets, even though there are extremely few examples in some classes, in which DRCW-ASEG can obtain significant improvement. This occurs, for example, in the datasets such as *Automobile* and *Cleveland*.
- (3) It is worth noting that in some majority classes, the proposed method DRCW-ASEG can improve the classification performance of concrete classes. For example, in the dataset *Lymphography*, where class 2 and class 3 are the majority classes, DRCW-ASEG obtains higher classification accuracy. The possible reason for this situation is that small disjuncts exist in multi-class imbalanced datasets.

6. Concluding remarks

In this paper, we have explored the relationship between multi-class imbalance learning and the non-competent classifier problem when the OVO decomposition scheme is used to solve

multi-class imbalanced datasets. Concretely, we have studied the synergy of dynamic classifier weighting in the OVO scheme in imbalanced scenarios. We have identified that the recent OVO approaches might have an additional bias towards majority classes during the decision making of the most competent classifiers.

On this account, we have proposed a new method called Distance-based Relative Competence Weighting with Adaptive Synthetic Example Generation (DRCW-ASEG). This novel algorithm dynamically explores the neighborhood of the instances to be classified, searching for imbalanced deficiencies. Learning from local information, DRCW-ASEG generates new synthetic examples, which are incorporated into the dynamic weighting stage process. We have tested the DRCW-ASEG approach versus the previous algorithm DRCW-OVO considering different binary imbalance learning methods. The achieved results reflect a great improvement of the classification performance using the DRCW-ASEG approach. With respect to the comparison of DRCW-ASEG and the state-of-the-art methods, the results obtained confirm that DRCW-ASEG outperforms the previous techniques in most datasets analyzed. Finally, we have demonstrated the suitability of the mechanism of DRCW-ASEG in recognizing the minority classes.

The proposal of this paper provides a direction for the future research when using the OVO strategy in handling multi-class imbalanced datasets. In the future, there are several works remaining to be addressed. Among them, we intend to introduce a more appropriate mechanism for generating synthetic examples into the proposed framework. Furthermore, it would be interesting to extend our strategy to the active field, such as extreme learning machine [57] and deep belief network [58]. As future work, we would also like to develop Big Data solutions [59] to handle multi-class imbalanced problems.

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Zhong-Liang Zhang received his Ph.D. degree in system engineering from the Northeastern University, China. He is currently a Ph.D. student at the Department of Computer Science and Artificial Intelligence, University of Granada, Spain. He is currently an Associate Researcher in the Department of Information Management and Information System, Hangzhou Dianzi University, China. His current research interests include data mining, imbalanced learning and evolutionary algorithms.



Xing-Gang Luo received a M.Sc. degree in mechanical design and manufacturing from the Northeastern University, China, and a Ph.D. degree in system engineering from the Northeastern University. His current research interests include product planning, data mining and service operation management. He is the author of more than 30 papers on peer-reviewed journals including *EJOR*, *IJPR*, *IJPE*, *C&IE*, *KBS*, *IEEE SMC*, *IEEE EM* etc.



Sergio González received the B.Sc. and M.Sc. Degrees in computer science from the University of Granada (UGR), Granada, Spain, in 2014 and 2015, where he is currently working toward the Ph.D. degree in the Department of Computer Science and Artificial Intelligence. His research interests include data mining, imbalanced learning, monotonic classification, data preprocessing and big data.



Salvador García received the M.Sc. and Ph.D. degrees in Computer Science from the University of Granada, Granada, Spain, in 2004 and 2008, respectively. He is currently an Associate Professor in the Department of Computer Science and Artificial Intelligence, University of Granada, Granada, Spain. Salvador García has published more than 70 papers in international journals (more than 50 in Q1), with more than 4500 citations, h-index 26, over 45 papers in international conference proceedings (data from Web of Science). He is a member of the editorial board of the "Information Fusion" (Elsevier), "Swarm and Evolutionary Computation" (Elsevier) and "AI Communications" (IOS Press) journals, and he is co-Editor in

Chief of the international journal "Progress in Artificial Intelligence" (Springer). He is a co-author of the book entitled "Data Preprocessing in Data Mining" published by Springer. His research interests include data science, data preprocessing, Big Data, evolutionary learning, Deep Learning, metaheuristics and biometrics. Dr. García has been given some awards and honors for his personal work or for his publications in and conferences, such as IFSA-EUSFLAT 2015 Best Application Paper Award and IDEAL 2015 Best Paper Award. He belongs to the list of the Highly Cited Researchers in the area of Computer Sciences (2014–2017): <http://highlycited.com/> (Clarivate Analytics).



Francisco Herrera (SM'15) received his M.Sc. in Mathematics in 1988 and Ph.D. in Mathematics in 1991, both from the University of Granada, Spain. He is currently a Professor in the Department of Computer Science and Artificial Intelligence at the University of Granada. He has been the supervisor of 41 Ph.D. students. He has published more than 350 journal papers, receiving more than 57,000 citations (Scholar Google, H-index 118). He is co-author of the books "Genetic Fuzzy Systems" (World Scientific, 2001) and "Data Preprocessing in Data Mining" (Springer, 2015), "The 2-tuple Linguistic Model. Computing with Words in Decision Making" (Springer, 2015), "Multilabel Classification. Problem analysis, metrics and techniques" (Springer, 2016), among others. He currently acts as Editor in Chief of the international journals "Information Fusion" (Elsevier) and "Progress in Artificial Intelligence" (Springer). He acts as editorial member of a dozen of journals. He received the following honors and awards: ECCAI Fellow 2009, IFSA Fellow 2013, 2010 Spanish National Award on Computer Science ARITMEL to the "Spanish Engineer on Computer Science", International Cajastur "Mamdani" Prize for Soft Computing (Fourth Edition, 2010), IEEE Transactions on Fuzzy System Outstanding 2008 and 2012 Paper Award (bestowed in 2011 and 2015, respectively), 2011 Lotfi A. Zadeh Prize Best paper Award (IFSA Association), 2013 AEPIA Award to a scientific career in Artificial Intelligence, 2014 XV Andalucía Research Prize Maimónides, 2017 Security Forum I + D + I Prize, and 2017 Andalucía Medal (by the regional government of Andalucía). He has been selected as a Highly Cited Researcher <http://highlycited.com/> (in the fields of Computer Science and Engineering, respectively, 2014 to present, Clarivate Analytics). His current research interests include among others, soft computing (including fuzzy modeling, evolutionary algorithms and deep learning), computing with words, information fusion and decision making, and data science (including data preprocessing, prediction and big data).